

```
In [205... import astropy.units as u
import astropy.constants as const
import numpy as np

h = const.h          # Planck constant
kB = const.k_B       # Boltzmann constant
c = const.c          # Speed of light
pi = np.pi
G = const.G
e = np.e
```

11.1

```
In [206... # b

sigma_0 = 36 * u.Lsun / ( u.pc**2)
h_r = 4 * u.kpc
R_0 = 8.5 * u.kpc

MilkwayBrightness = sigma_0 / e**(-R_0/h_r)
MilkwayBrightness
```

Out[206... 301.42431 $\frac{L_{\odot}}{\text{pc}^2}$

```
In [207... # c
h_r = 4 * u.kpc

asinf = 2 * pi * MilkwayBrightness * h_r**2

print(f"asinf = {asinf.to(u.Lsun).value:.3e} {asinf.to(u.Lsun).unit}")

asinf = 3.030e+10 solLum
```

```
In [208... mwMass = asinf/3 / u.Lsun * u.Msun

print(f"mwMass = {mwMass.to(u.Msun).value:.3e} {mwMass.to(u.Msun).unit}")

mwMass = 1.010e+10 solMass
```

In []:

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In [209... # e

h_r = 4 * u.kpc
R_0 = 8.5 * u.kpc

R_0Lum = 2 * pi * MilkwayBrightness * h_r**2 * (1 - (((R_0/(h_r) + 1)))

print(f"R_0Lum = {R_0Lum.to(u.Lsun).value:.3e} {R_0Lum.to(u.Lsun).unit}")

R_0Lum = 1.899e+10 solLum
```

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In [210... R_0Lum/asinf
```

Out[210... 0.62677197

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In [211... (1 - ((R_0/h_r) + 1)) * e **(-R_0/h_r))
```

```
Out[211... 0.62677197
```

11.2

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In [212... # a
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```
v_esc = 500 * u.km / u.s
R = 17/2 * u.kpc

M = ((v_esc * np.sqrt(R))/np.sqrt(2 * G))**2

print(f"M = {M.decompose().value:.3e} {M.decompose().unit}")

print(f"M = {M.to(u.Msun).value:.3e} {M.to(u.Msun).unit}")
```

```
M = 4.912e+41 kg
M = 2.470e+11 solMass
```

```
In [213... # b
```

```
m_v = 18.9
P = 31.47 * u.day

M_v = -2.76 * np.log10(P/(10 * u.day)) - 4.16
print(f"M_v = {M_v.decompose().value:.3e} {M_v.decompose().unit}")

d = 10**((m_v - M_v + 5)/5) * u.pc

print(f"d = {d.decompose().value:.3e} {d.decompose().unit}")
print(f"d = {d.to(u.Mpc).value:.3e} {d.to(u.Mpc).unit}")

# d.decompose()
```

```
M_v = -5.534e+00
d = 2.378e+22 m
d = 7.706e-01 Mpc
```

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In [214... # B ii
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H_0 = 70 * (u.km / u.s) / u.Mpc
rec_v = d * H_0
rec_v.to(u.km / u.s)
```

```
Out[214... 53.943242  $\frac{\text{km}}{\text{s}}$ 
```

```
In [215... # B iii
```

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c_v = -119 * u.km / u.s
delta_v = c_v - rec_v
delta_t = abs(14*u.Gyr)

a = delta_v / delta_t
print(f"a = {a.decompose().value:.3e} {a.decompose().unit}")
```

```
a = -3.914e-13 m / s2
```

```
In [216... # C iv

R = d

M_MW = -1 * (a * R**2/(3*G))

print(f"M_MW = {M_MW.to(u.Msun).value:.3e} {M_MW.to(u.Msun).unit}")

M_MW = 5.559e+11 solMass
```

```
In [ ]:
```

11.3 Black Hole Masses

```
In [217... # A
# R=Dθ
d = 16.8 * u.Mpc
theta = .1 * u.arcsec
r = d*theta/u.rad
r.to(u.pc)
```

```
Out[217... 8.1448698 pc
```

```
In [218... # B

pluspointlarc = 1780 * u.km /u.s
minuspointlarc = 650 * u.km /u.s
c_vel = (pluspointlarc + minuspointlarc)/2
c_vel
```

```
Out[218... 1215  $\frac{\text{km}}{\text{s}}$ 
```

```
In [219... # C
v = 350 * u.km /u.s
R = r
M = (c_vel**2 * R)/G

print(f"M = {M.decompose().value:.3e} {M.decompose().unit}")
print(f"M = {M.to(u.Msun).value:.3e} {M.to(u.Msun).unit}")
```

```
M = 5.559e+39 kg
M = 2.796e+09 solMass
```

11.4 Redshifts and the Hubble Law

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In [220... # a

restWv = 486.135 *u.nm
obsWv = 497.97 * u.nm

z = (obsWv - restWv)/obsWv
z
```

```
Out[220... 0.023766492
```

```
In [221... # b
H_0
5 * np.log10(c/H_0 * 10**5 / u.Mpc)
```

Out[221... 43.158613

```
In [222... (c/H_0 * 10**5 / u.Mpc).decompose()
```

Out[222... 4.2827494×10^8

```
In [226... # c

Z = .023
m_v = 12.7

M = -(5*np.log10(z) + 43.2 - m_v)
M
```

Out[226... -22.379825

11.5

```
In [224... gal_Distnace = (2 * 10**6) / (5.21553 * 10**(20)) * u.Mpc
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```
In [225... recVel = (H_0 * gal_Distnace)

print(f"recVel = {recVel.to(u.km/u.s).value:.3e} {recVel.to(u.km/u.s).uni

recVel = 2.684e-13 km / s
```